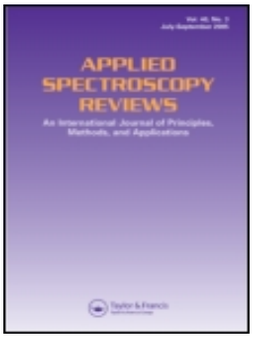

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Acquisition of a series of temperature-varied sample spectra to induce characteristic structural changes of components and selection of target-descriptive variables among them for multivariate analysis to improve accuracy

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ABSTRACT

As a means of improving the accuracy of Raman spectroscopic quantitative analysis, a strategy combining the generation of a series of temperature-varied spectra to make diverse and characteristic structural information of sample components widely available for calibration, and subsequent selection of more property-descriptive variables among these spectra, has been demonstrated. For the evaluation, Raman spectra of synthetic hydrocarbon mixtures, lube base oils (LBOs) and polyethylene (PE) pellets were acquired at regular intervals while the sample temperature gradually increased from cryogenic to near room temperature. To select target-descriptive variables from all of the snapshot (temperature-varied) spectra, a Markov blanket (MB) feature (variable) selection able to produce a minimal set of features without changing the original target distribution was adopted. The selection utilizes a conditional independence test to quickly obtain an optimal feature subset by simultaneously considering relevance to a target variable and redundancy between selected features without using any heuristic searching. When MB-selected variables were used for partial least squares (PLS) to determine the concentrations of components in the hydrocarbon mixtures, kinematic viscosity of 40°C (KV@40) of LBOs, and density of PE pellets, the accuracy was improved compared to the use of either all snapshot spectra without variable selection or an optimal single snapshot spectrum. The incorporation of component-specific and property-descriptive variables without redundant information for PLS was the origin of the improvement in accuracy. The proposed method could potentially be extended to the analysis of other complex samples including petroleum-driven samples, edible oils, and other polymers.

KEYWORDS

Temperature-varied Raman spectra; variable selection; Markov blanket; lube base oils; polyethylene pellets

Introduction

We have previously demonstrated the strategy of improving accuracy for multivariate quantitative analysis of samples by acquiring their Raman spectra at an optimal temperature

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below the zero range, where spectral discrimination among sample components was enhanced and more sample-descriptive spectral features were provided (1–3). Based on the fact that molecular vibrations of a component at a certain temperature are relatively more distinct than those of other components, a subsequently obtained spectrum could be more informative for building a multivariate correlation with either the concentration of a component or a sample property (4–8). The effectiveness of this strategy was demonstrated for three different Raman spectroscopic analyses: the compositional analysis of synthetic hydrocarbon mixtures, determination of viscosity of LBOs, and measurement of density of PE pellets. In all of these cases, accuracies for the determination of either component concentrations or physical properties of the samples were improved, when Raman spectra acquired at optimal temperatures were used for PLS (9–11).

Although the spectrum acquired at an optimal snapshot (temperature) provided more target-descriptive information for multivariate analysis, only one snapshot spectrum out of all of the snapshot spectra (continuously collected at regular intervals during temperature elevation from cryogenic to nearly room temperature) was utilized, while different vibrational information contained in the rest of the spectra could also be valuable for the analysis. Therefore, it is worthwhile to investigate the possibility of further improving the accuracy by fully utilizing information from all snapshot spectra, equivalently incorporating all temperature-induced changes in molecular vibrations of components. When all of the snapshot spectra are used for multivariate analysis, two choices are available. The first option is an incorporation of all snapshot spectra for multivariate analysis. This option is advantageous since it utilizes all of the information present in the snapshot spectra for calibration; however, the substantial increase in the number of dimensions (variables) make the necessary data sparse; this is the so-called curse of dimensionality problem, and it negatively impacts the accuracy of a multivariate analysis. In this situation, model prediction accuracy may be degraded if less relevant variables to a target variable are incorporated in the calibration, especially with a finite number of observations (samples). The second choice is the use of properly selected variables out of all snapshot spectra using feature selection methods. Successful feature selection could improve accuracy by incorporating only target-descriptive information without adding redundant and unnecessary features into the model. In this way, the high-dimensionality problem can be also easily remedied.

In this study, a strategy of combining the generation of a series of temperature-varied spectra to make diverse structural information of sample components widely available for calibration, and the subsequent selection of more property-descriptive variables among the spectra, was demonstrated. For the evaluation, all snapshot Raman spectra of synthetic hydrocarbon mixtures (1), LBOs (2), and PE pellets (3) acquired in the previous studies were again employed. In each case, diverse temperature-induced spectral variations, such as changes in bandwidth and position, and the appearance of new bands were present in the snapshot spectra. A series of snapshot spectra was sequentially integrated as a single spectral dataset for further analysis.

For the selection of variables (features) from the entire data set, both filter and wrapper methods are commonly available (12, 13). A filter method defines a feature importance metric to obtain important features by ordering the metric. In addition, feature selection and subsequent multivariate modeling are separately performed. On the other hand, a wrapper method, such as stepwise regression, iteratively selects features in conjunction with modeling to ultimately improve accuracy. Filter methods generally provide less accurate results than wrapper methods, but they are much simpler and faster. When a dataset possesses a large

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number of features compared to that of samples, wrapper methods are less preferred due to the enormous number of computations (14). Thus, filter methods were more appropriate for this study, in which the snapshot-integrated data contained up to 417,060 variables.

Filter methods are divided into two categories of feature ranking (FR) and feature subset selection (FSS) (13, 15). FR methods, such as regressional Relief-F (16), information gain (17, 18), and Chi-square (19) based on pairwise statistics, tend to select features that are individually relevant for a target property without considering their interactions (correlation) (20). Redundant features are an unwanted latent source potentially degrading accuracy of multivariate analyses. In addition, determination of an optimal number of selected variables is tricky in practice. On the other hand, one of the FSS methods, correlation-based feature selection (CFS) (21) searches a sub-optimal subset of variables by considering their relevance-redundancy tradeoff. CFS utilizes a heuristic search strategy such as genetic search (22) and best-first search (23, 24); however, results of CFS could sensitively vary depending on the employed search method, which is undesirable in multivariate analysis.

Markov blanket (MB) (25, 26) feature selection can address the above-mentioned problem in CFS. MB was theoretically proven to produce a minimal set of features without changing the original target distribution (27). MB feature selection utilizes a conditional independence (CI) test to quickly obtain an optimal feature subset, while simultaneously considering relevance to a target variable and redundancy between selected features without using any heuristic searches. It is simple to implement because only one parameter, a significance level of the CI test, is involved as a selection criterion. Several studies have shown the superior performance of MB when handling datasets composed of a small number of samples with a large number of corresponding variables (28, 29). The usefulness of MB has been also demonstrated in the field of machine learning, especially for bioinformatics such as analysis of genomic microarray data and mass spectra of diverse biological samples (30–32).

Here, an incremental association Markov blanket (IAMB) algorithm (33), which initially finds relevant variables to a target property and then reduces the selected MB variables using CI tests, has been employed to select target-descriptive variables in the integrated dataset; the selected variables were subsequently used for PLS to determine the concentrations of components in hydrocarbon mixtures, kinematic viscosity at 40°C (KV@40) of LBOs, and density of PE pellets as shown in Table 1. The integrated datasets consisted of a huge number of features (variables, intensities at each wavenumber), while the number of observations (samples) was much smaller such as less than 28 samples. In each case, PLS accuracies resulting from the use of either whole variables or IAMB-selected variables were compared along with those achieved using only one optimal snapshot spectrum as reported in the previous studies (1–3). The characteristics of IAMB, such as which variables were selected in the sample spectra and which snapshot spectra contained them, were then examined. Finally, the potential of the proposed method in multivariate spectroscopic analyses has been discussed.

Experimental

Preparation of Raman spectral datasets

Raman spectral datasets of synthetic hydrocarbon mixtures (25 samples), LBOs (28 samples), and PE pellets (25 samples) acquired in the previous studies were used (1–3). Briefly, a glass vial containing a sample was initially immersed in liquid nitrogen for sufficient cooling, and then the

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Table 1. SEPs resulting from concentration determination of 11 different components in a hydrocarbon mixture, KV@40 of LBO samples, and density of PE pellets with the use of all snapshot spectra (whole variables), IAMB-selected variables, and the optimal snapshot spectrum. The numbers in parentheses indicate the corresponding optimal snapshot spectrum number.

			Whole variables			IAMB-selected variables			Optimal snapshot spectra		
Sample	Component or property	Unit	# of variables	# of factors	CV-SEP	# of variables	# of factors	CV-SEP	# of variables	# of factors	CV-SEP
Hydrocarbon mixture	*n*-Hexane	wt%	417,060	16	1.57	39	8	0.31	4,634 (4)	14	0.41
	n-Heptane	wt%	417,060	14	1.50	42	5	0.26	4,634 (7)	13	0.39
	n-Octane	wt%	417,060	16	1.53	31	4	0.37	4,634 (7)	10	0.56
	n-Nonane	wt%	417,060	14	1.66	36	7	0.44	4,634 (5)	12	0.47
	Isooctane	wt%	417,060	16	0.78	35	7	0.10	4,634 (10)	14	0.35
	Cyclohexane	wt%	417,060	8	0.33	41	10	0.40	4,634 (7)	5	0.35
	Methylcyclohexane	wt%	417,060	15	0.53	41	6	0.22	4,634 (7)	12	0.23
	Benzene	wt%	417,060	7	0.20	44	6	0.43	4,634 (25)	4	0.36
	Toluene	wt%	417,060	6	0.34	44	4	0.58	4,634 (37)	6	0.67
	p-Xylene	wt%	417,060	8	0.40	40	5	0.35	4,634 (48)	9	0.47
	Indan	wt%	417,060	13	0.41	41	9	0.20	4,634 (7)	12	0.22
LBO	KV@40	cSt	243,090	7	2.15	66	5	1.50	2,701 (7)	4	1.97
PE pellet	Density	g cm^{-3}	30,962	7	0.00074	23	5	0.00058	1,550 (7)	6	0.00069

sample vial was quickly removed for continuous Raman spectral collection during natural temperature elevation to near room temperature. During the temperature elevation, 90, 90, and 25 snapshot spectra (resolution: 4 cm^{-1}) were collected for the hydrocarbon mixture, LBOs, and PE pellets, respectively. Triplicate spectra were collected for each sample by repeating this procedure. Each snapshot spectrum contained 4634, 2701, and 1550 variables and thus, the total numbers of variables obtained by combining all of the snapshot spectra in the datasets of the hydrocarbon mixture, LBOs, and PE pellets were 417,060, 243,090, and 38,750, respectively. Using either whole snapshot spectra or selected variables from them, the concentrations of each component in the hydrocarbon mixture, KV@40 of LBOs, and density of PE pellets (measured by a standard ASTM method at 23°C) were determined using PLS.

Procedure for IAMB-based variable selection

The concept of the MB comes from a Bayesian network that shows a causal relationship among variables via a directed acyclic graph. The nodes of a Bayesian network represent feature, and its edges represent conditional dependencies between nodes. From the perspective of probability, the MB of a target variable T, denoted by $MB(T)$, is the minimal set of variables conditioned on which all other variables are independent of T. From the graphical perspective as shown in Figure 1, the MB of T is the union of the parent child (PC) nodes and spouse (SP) node of T. For example, PC nodes of T are $PC(T) = \{\lambda_2, \lambda_3, \lambda_6\}$, and SP is $SP(T) = \{\lambda_5\}$. So, MB of T is $MB(T) = \{\lambda_2, \lambda_3, \lambda_5, \lambda_6\}$. It means that nodes λ_1, λ_4 and λ_7 are independent of T conditioned on $MB(T)$ (34). In the case of spectroscopic data, T and $\lambda_1, \ldots, \lambda_7$ in the figure are a property of a sample (such as component concentration or physical property) to be measured and intensities at individual wavelengths, respectively.

Figure 2 shows the overall procedure for IAMB-based variable selection. The pseudo code of IAMB is present in the supplementary information for reference. As shown, IAMB utilizes

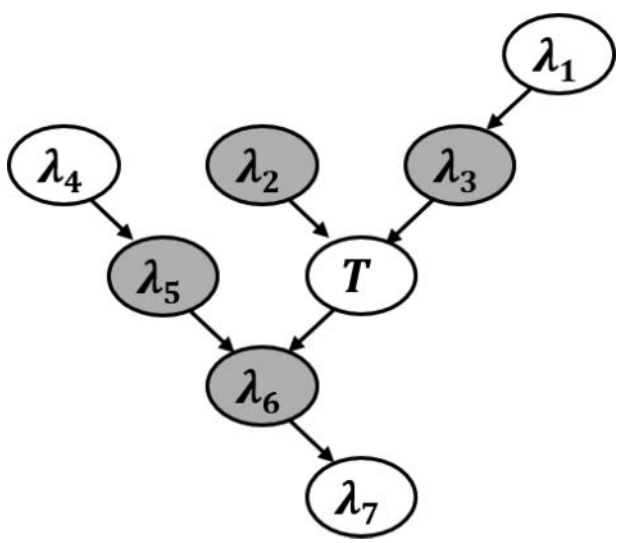

Figure 1. Graphical description of a set of MB variables chosen for a target variable (T). Nodes (ovals) and edges (arrows) correspond to random variables and conditional dependencies, respectively. The shaded ovals are parent child (PC) and spouse (SP) nodes.

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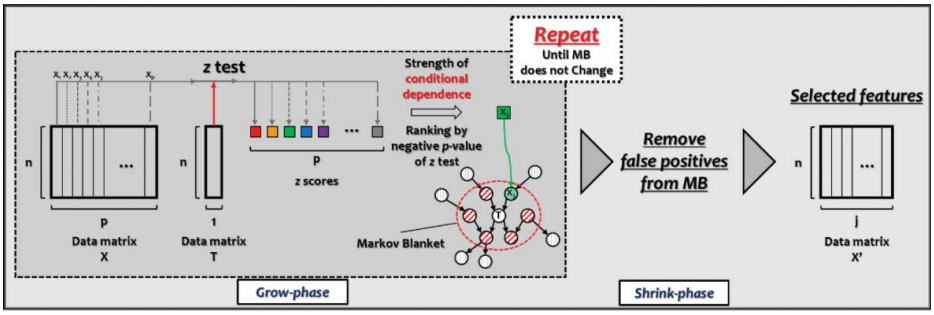

Figure 2. Graphical presentation of the overall procedure for IAMB-based variable selection. IAMB utilizes a grow-and-shrink approach composed of two phases (grow and shrink phases) in variable selection as described.

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a grow-and-shrink approach composed of two phases for variable selection. In the grow-phase (lines 2–6 in the pseudo code), variables determined to be conditionally dependent on a target variable are added to the MB through the CI test based on Fisher's *z*-test (refer to supplementary information for a detailed description). It is based on partial correlation which measures the degree of association between two random variables. All variables are initially ranked in order of their strength of dependency with the target variable. The strength of dependency is estimated by the *p* value of the *z*-test. The most relevant variable, the one possessing the smallest *p* value, out of all of the variables is selected as a first candidate for the MB (such as X_3 of the green box). Then, the variable will be included into the MB if it is determined to be dependent on the target variable by hypothesis testing. Next, variables other than the selected one in the MB are evaluated by the ranking of their strength of conditional dependency with the target variable, and the second variable is determined in the same fashion as described above. This procedure is repeated until no further variable is added into the MB.

In the shrink-phase (lines 7–9 in the pseudo code), among the selected variables in the MB, variables conditionally independent of the target variable are removed. Specifically, each variable is sequentially considered in the same order as in the previous growing-phase, and then it is determined whether the variable is dependent on the target variable or not, when the rest of the features in the MB are conditioned. When a variable is found to be conditionally independent, it is regarded as a false positive of the MB formed in the grow-phase and is thus removed from the MB. This step continues until all of the selected variables in the MB are individually screened as described above. The variables remaining in the MB after the shrink-phase are finally used for subsequent multivariate analysis. PLS and IAMB were accomplished using the MATLAB R2014a (The MathWorks Inc.) software and MATLAB version of Causal Explorer toolkit (35), respectively.

Results and discussion

Comparison of PLS results for compositional analysis of hydrocarbon mixtures

Table 1 shows the standard error of predictions (SEPs) resulted from the concentration determination of 11 different components in the hydrocarbon mixture when all snapshot spectra, as well as IAMB-selected variables, were employed for PLS. The number of

IAMB-selected variables in each case is indicated. SEPs obtained using only the optimal snapshot spectrum for each component, as presented in the previous report (1), are shown for comparison. The numbers in parentheses indicate the corresponding optimal snapshot spectrum number. In all cases, SEPs were obtained by five-fold cross-validation (CV) with 1,000 runs. For calibration, 80% of the total samples were randomly selected, and the remaining samples were used for validation. Figure 3 and Figure 4 show the resulting concentration correlation plots for the concentration determination of each component when all snapshot spectra (left column), IAMB-selected variables (middle column), and the optimal single snapshot spectrum (right column) were used for PLS.

In measurements of *n*-hexane, *n*-heptane, *n*-octane, and *n*-nonane (*n*-paraffins), when all snapshot spectra were used together for PLS, the errors were considerably higher than those obtained using either IAMB-selected variables or optimal snapshot spectra. The molecular structures of these *n*-paraffins are very similar to each other, thus their temperature-dependent structural variations are accordingly analogous. This means that their bands are fairly overlapped throughout all of the snapshot spectra, although a component may be relatively more discernible from other components at a certain temperature. In this situation, it would be difficult to find minor spectral differences among the components in the all-snapshot integrated spectra with extremely high dimensions, and substantial degradation of accuracy is expected. When only the optimal snapshot spectrum was used in each case, the SEPs decreased dramatically, which was attributed to the use of more target-component discriminable features in the corresponding optimal snapshot spectrum containing much lower dimensions for modeling.

When IAMB-selected variables were used for the determination of *n*-paraffin concentrations, the SEPs were further improved in each case compared with those obtained using the optimal snapshot spectrum. Only 31 to 42 variables selected out of total 417,060 variables were more descriptive to follow the concentration variations of *n*-paraffins by employing an even smaller numbers of factors. For detailed examination, the locations of selected variables in the sample spectra for the determination of each component concentration were analyzed. Figure 5(a) shows the wavenumbers (locations) of IAMB-selected variables in the sample spectrum (top) and the snapshot spectrum numbers containing the variables for the determination of *n*-hexane concentrations (bottom). In addition, actual measurement temperatures corresponding to the snapshot numbers are indicated on the right *y*-axis. All 90 snapshot spectra of the sample containing the highest concentration of *n*-hexane are displayed. On the spectra, the positions of selected variables are indicated in red. The inset highlights the 994 to 339 cm^{-1} range containing many peaks with low intensities, in which yellow highlights are used to indicate the selected variables in the range. A notable observation was that the selected variables were fairly spread across the spectral range as well as the snapshots, rather than localized on certain bands. Figure 7(a) shows the number of variables selected from each snapshot spectrum for the determination of the *n*-hexane concentration. The measurement temperatures of the corresponding snapshot spectra are indicated on the top *x*-axis. Although five variables were selected from the snapshot #1 spectrum, only one to three variables were chosen from each selected snapshot spectrum. Again, the selection of a large number of variables from a certain snapshot spectrum was not observed. The overall observations shown in the figures demonstrate that more *n*-hexane selective spectral features compared to the other components are widely present throughout the snapshot spectra, and the selection of more descriptive variables available in all snapshot spectra for PLS eventually

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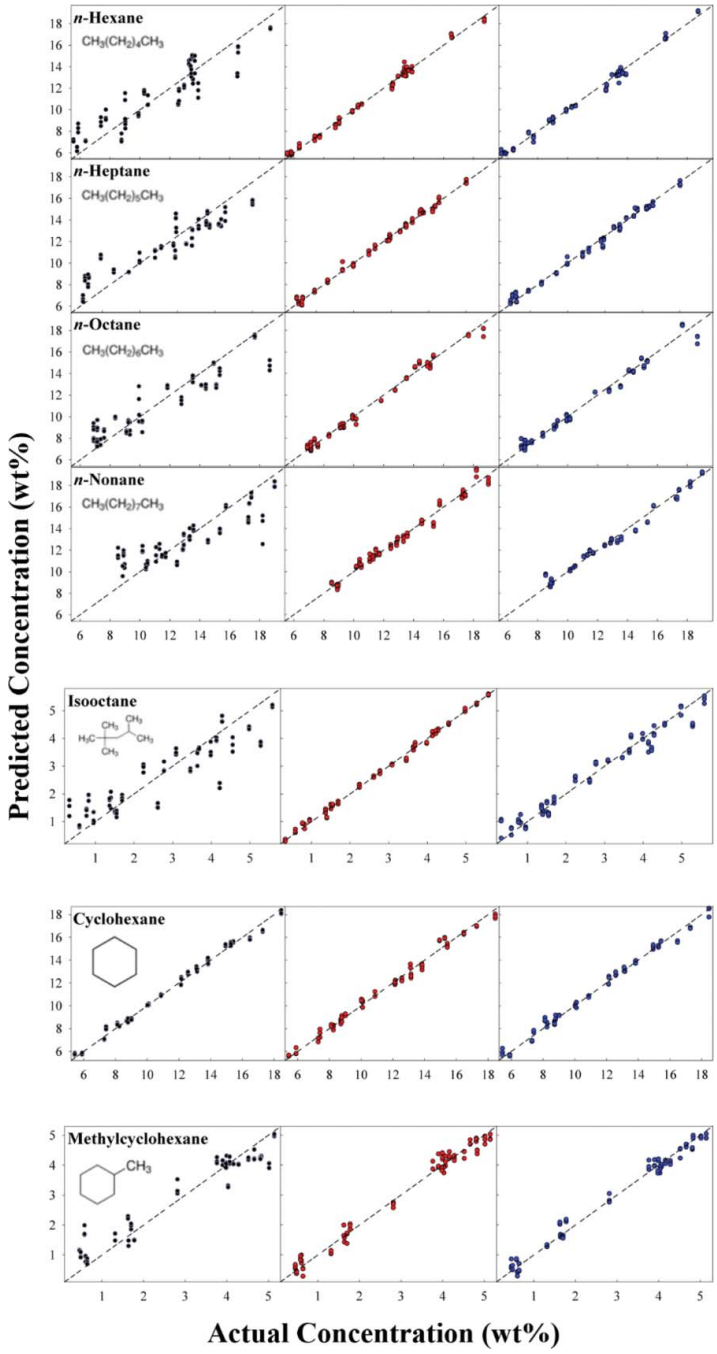

Figure 3. Concentration correlation plot results for the concentration determination of *n*-paraffin and naphthene components when all of the snapshot spectra (left column), IAMB-selected variables (middle column), and the optimal single snapshot spectrum (right column) are used for PLS.

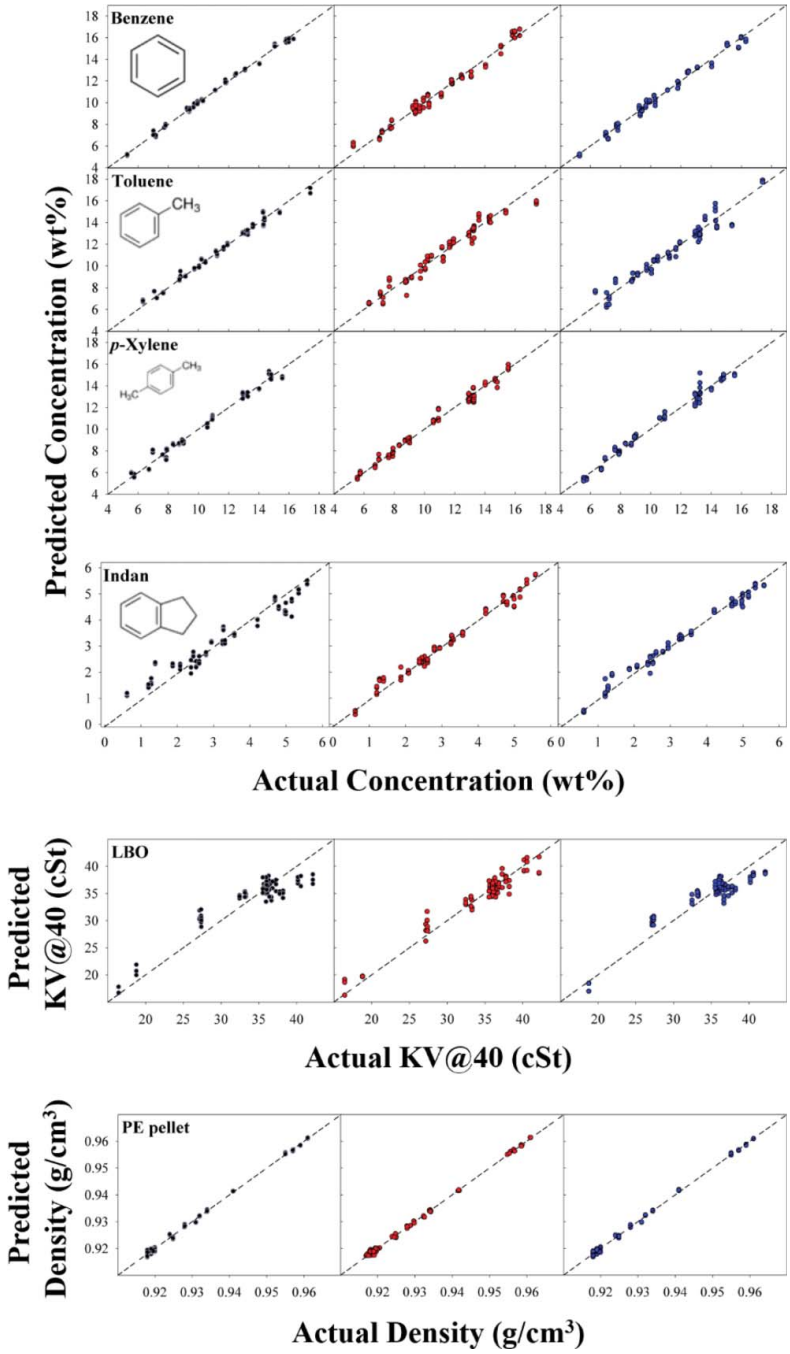

Figure 4. Concentration (property) correlation plot results for determinations of concentrations of aromatic components in a hydrocarbon mixture, KV@40 of LBOs, and density of PE pellets, when all of the snapshot spectra (left column), IAMB-selected variables (middle column), and the optimal single snapshot spectrum (right column) are used for PLS.

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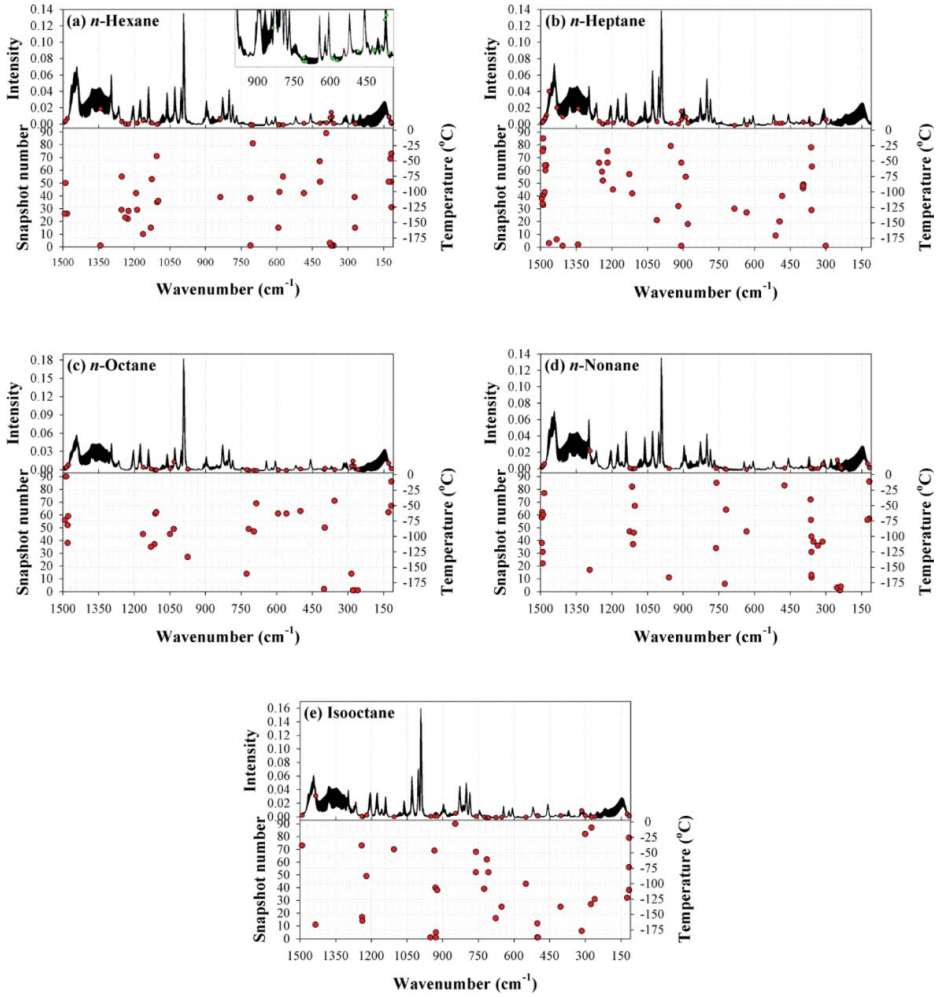

Figure 5. The locations of selected variables in the sample spectra and snapshot spectrum numbers containing the corresponding variables for the determination of *n*-hexane (a), *n*-heptane (b), *n*-octane (c), *n*-nonane (d), and isooctane (e) concentrations. In each case, all 90 snapshot spectra of the sample containing the highest concentration of the corresponding component are also displayed.

leads to a more accurate concentration determination. The spectral features in the snapshot #4 spectrum are relatively more descriptive to correlate with the concentration variation compared to those in the rest of the snapshot spectra, as previously observed (1).

Figures 5(b), (c), and (d) show the locations of the selected variables in the sample spectra and the snapshot numbers containing the variables for the determination of *n*-heptane, *n*-octane, and *n*-nonane concentrations, respectively (bottom). In each case, all 90 snapshot spectra of the sample containing the highest concentration of the corresponding component are also displayed (top). Figures 7(b), (c), and (d) show the number of selected variables in each snapshot spectrum employed for the determination of *n*-heptane, *n*-octane, and *n*-nonane concentrations, respectively. Again, the selected variables are fairly spread along the spectral range as well as the snapshots, and the incorporation of IAMB-selected variables for PLS decreased the SEPs in all three cases. As found in the previous report (1), *n*-paraffins

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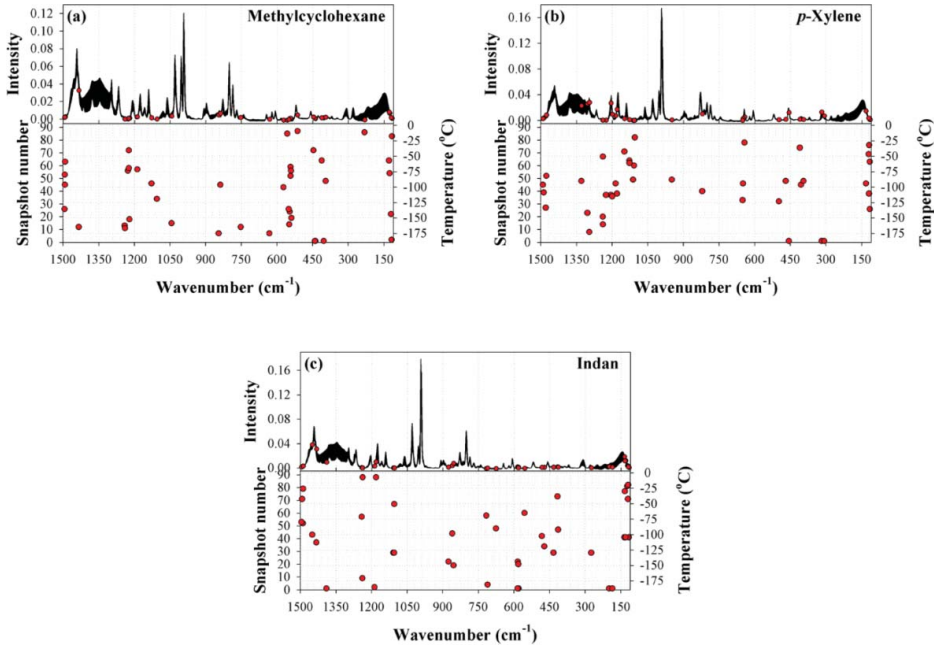

Figure 6. The locations of selected variables in the sample spectra and snapshot spectrum numbers containing the corresponding variables for the determination of (a) methylcyclohexane (a), *p*-xylene (b), and indan (c) concentrations. In each case, all 90 snapshot spectra of the sample containing the highest concentration of the corresponding component are also displayed.

are easily compressible and deformable under the temperature change from cryogenic to room temperature. Therefore, diverse temperature-induced spectral variations are present throughout the snapshot spectra, and the selection of more component-specific information with less spectral overlapping with other components of similar molecular structure is advantageous for improving the accuracy in the determination of their concentrations (refer to Figure 3). In parallel, the results obtained for the determination of isooctane (2,2,4-trimethylpentane, *iso*-paraffin) concentration (refer to Figure 5(e) and Figure 7(e)) were similar, and the improvement of accuracy was significant with the use of IAMB-selected variables (refer to Figure 3).

In contrast to the above observations, the accuracies in the measurements of cyclohexane, benzene, and toluene were better when all of the snapshot spectra were employed, rather than the use of either optimal snapshot spectra or IAMB-selected variables (refer to Table 1, Figure 3 and Figure 4). The vibrations of the saturated ring of cyclohexane and the benzene ring are relatively simple and characteristic. In addition, as shown in the previous work (1), the differences between their spectral features acquired in the 1st snapshot and at room temperature were relatively insignificant due to the rigid and hardly deformable ring structures. Therefore, their presence in the sample mixture is easily recognizable throughout the snapshot spectra, and the use of all snapshot spectra provides clearly descriptive spectral features into PLS to correlate with their concentration variations. The use of a single optimal spectrum or a much smaller number of IAMB-selected variables would partially miss available component-specific information present throughout the snapshot spectra, and the resulting accuracies would expect to be slightly inferior. In the case of toluene, a methyl group

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attached to a benzene ring would produce a more diverse temperature-dependent spectral variation, while the structure of toluene is still distinctive due to its benzene ring. Thus, recognizable toluene-specific information exists extensively over the snapshot spectra.

For the remaining components (methylcyclohexane, *p*-xylene, and indan) possessing less distinct molecular structures compared to cyclohexane and benzene, the measurements were most accurate when IAMB-selected variables were used in each case (refer to Table 1, Figure 3 and Figure 4). The relatively more complex molecular structures lead to greater overlapping of spectral features with those of other components such as *n*-paraffins, and thus the selection of more component-specific information is beneficial to follow the corresponding concentration variations more accurately. Figures 6(a), (b), and (c) show the locations of selected variables in the sample spectra and snapshot spectrum numbers containing the corresponding variables for the determination of methylcyclohexane, *p*-xylene, and indan concentrations, respectively. All 90 snapshot spectra of the sample containing the highest concentration of the corresponding component are also displayed for each case. Figures 7(f), (g), and (h) show the number of selected variables in each snapshot spectrum used for the measurements of methylcyclohexane, *p*-xylene, and indan, respectively. The selected variables, as well as the snapshots, are widely spread across the spectral range, as also shown in the measurements of the *n*-paraffins and isooctane.

Overall, IAMB was effective in finding component-specific information present in all of the snapshot spectra, especially for components with similar molecular structures, such as *n*-paraffins and *iso*-paraffins, and the use of IAMB-selected variables for PLS was helpful in improving the accuracies for determination of their concentrations. Meanwhile, measurements of components possessing distinct molecular structures relative to those of other components, such as cyclohexane and benzene, were more accurate when all of the snapshot spectra were utilized, as their distinct component-specific information is broadly present throughout the snapshots.

Comparison of PLS results for the determination of KV@40 of LBOs and density of PE pellets

LBOs are highly complex mixtures predominantly containing long-chain (C_{16}–C_{25}) *n*-paraffins, and *iso*-paraffins. To determine the KV@40, a bulk physical property resulting from the contributions of individual components as well as their intermolecular interactions, better spectral discrimination among the components in LBO samples is beneficial in building a more accurate multivariate correlation with the change of KV@40. The potential improvement of accuracy by utilizing more KV@40-descriptive variables out of all snapshot spectra is also of great interest. Table 1 and Figure 4 show SEPs and the corresponding property-correlation plots, respectively, resulting from the determination of KV@40 of LBO samples with the use of all snapshot spectra, IAMB-selected variables, and the optimal snapshot spectrum determined in the previous publication (2). As shown, the use of IAMB-selected variables substantially improved the accuracy. The number of selected variables was 66, which is larger than the number of variables used in the compositional analysis of hydrocarbon mixtures. Since the spectral features of LBO are much more complex and have greater overlap due to the composition of long-chain hydrocarbons with diverse molecular structures, the employment of a greater number of variables is necessary to describe the variation of KV@40.

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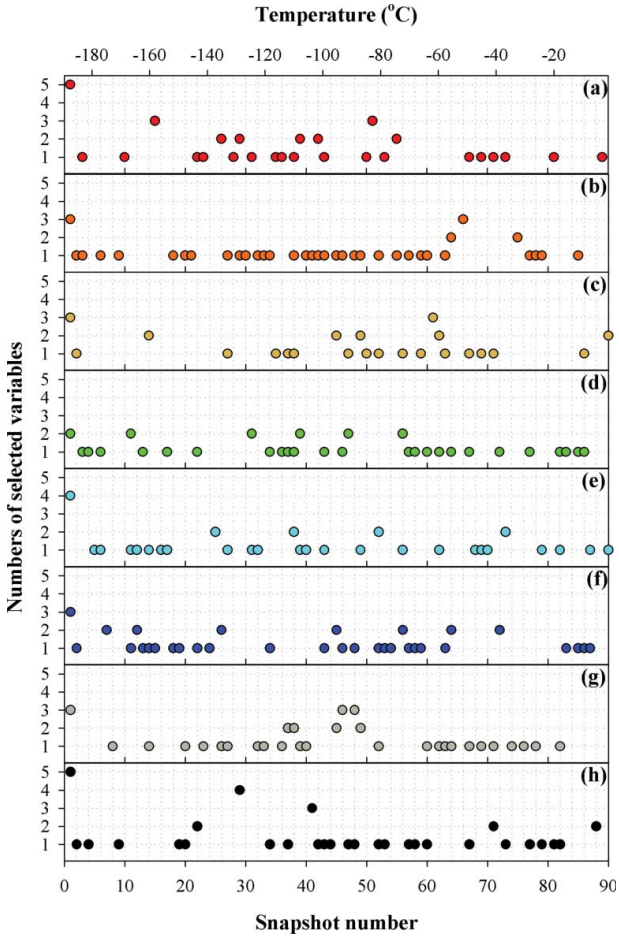

Figure 7. The number of selected variables from each snapshot spectrum for the determination of (a) *n*-hexane, (b) *n*-heptane, (c) *n*-octane, (d) *n*-nonane, (e) isooctane, (h) methylcyclohexane, (g) *p*-xylene, and (h) indan concentrations.

Figure 8(a) shows the locations of selected variables in the sample spectra (top) and snapshot spectrum numbers containing the corresponding variables for the measurement of KV@40 of the LBO samples (bottom); all snapshot spectra of an LBO sample with a middle KV@40 value (0.5 cSt) are displayed, with locations of the selected variables indicated by red circles on the spectra. Figure 8(b) shows the number of variables selected from each snapshot spectrum for the determination of KV@40. As shown in the figures, the selected variables were again widely distributed over the spectral range without localization on a certain range of snapshots (temperature range). These results suggest that diverse and useful KV@40-descriptive information is present in all of the snapshot spectra, and the use of descriptive variables properly selected from them for PLS drives the improvement of accuracy.

As the temperature of a polymer was changed from the sub-zero to cryogenic range, large spectral variations originating from temperature-dependent changes of the polymeric structure and phase transitions were observed, as shown in the former study (3).

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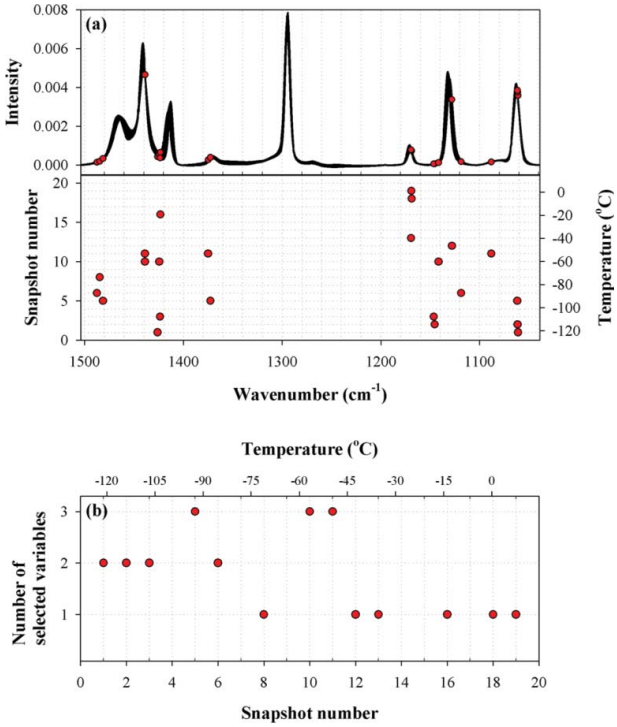

Figure 8. The locations of selected variables in the sample spectra and snapshot spectrum numbers containing the corresponding variables for the measurement of KV@40 of LBO samples (a). All snapshot spectra of an LBO sample with a middle KV@40 value (0.5 cSt) are displayed, in which locations of the selected variable are indicated with red circles on the spectra. The number of variables selected from each snapshot spectrum is also shown (b).

Thus, it is again of great interest to evaluate the possibility of improving the multivariate correlation between polymer properties and the corresponding spectral features present in all of the temperature-induced polymer spectra. Table 1 and Figure 4 show SEPs and the corresponding property-correlation plots, respectively, resulting from the determination of the density of PE pellets when all of the snapshot spectra, IAMB-selected variables, and the optimal snapshot spectrum were used. Again, employment of IAMB-selected variables (23 variables) for PLS provided the most accurate correlation with the densities of PE pellets.

Figure 9(a) shows the locations of selected variables in the sample spectra (top) and snapshot spectrum numbers containing the corresponding variables for the determination of PE pellet densities (bottom). All snapshot spectra of a low density PE pellet sample (0.9112 g cm^{-3}) are shown with the locations of the selected variables indicated in red. Figure 9(b) shows the number of variables selected from each snapshot spectrum for the measurement. The variables in the 1500 to 1350 cm^{-1} and the 1200 to 1050 cm^{-1} ranges were selected, while they were not localized on a certain range of snapshots. Spectral information descriptive of density variation was widely present in the series of spectra collected with the temperature variation, and the selection of density-descriptive variables from them is useful in more accurately following the change of PE density, a physical property.

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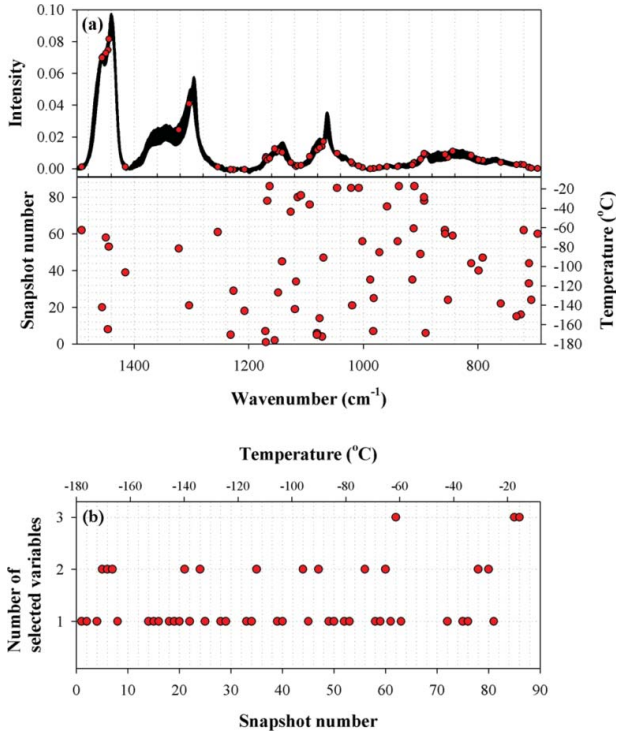

Figure 9. The locations of selected variables in the sample spectra and snapshot spectrum numbers containing the corresponding variables for the determination of PE pellet densities. All snapshot spectra of a low-density PE pellet sample (0.9112 g cm^{-3}) are displayed with red highlights indicating the locations of the selected variables are shown (a). The number of variables selected from each snapshot spectrum is also shown (b).

Conclusion

The IAMB-based selection of variables out of all of the snapshot spectra that contain diverse temperature-induced structural variations, and the use of the selected variables for PLS, were effective in improving accuracies in the determination of component concentrations in a synthetic hydrocarbon mixture, KV@40s of LBOs, and densities of PE pellets, compared to the use of either all of the snapshot spectra without variable selection or an optimal single snapshot spectrum. In the analysis of the hydrocarbon mixture, the use of all snapshot spectra provided more accurate results for the measurements of components with distinct molecular structures (cyclohexane, benzene, and toluene), since component-specific information was present in all of the snapshot spectra due to their distinct molecular structures. The overall results support that the proposed scheme is more versatile when concentrations of components with similar molecular structures are measured or physical properties of highly complex samples are determined. We believe that this method would be further applicable to the analysis of many other complex samples, for example petroleum-based samples (such as naphtha, gasoline, and kerosene), edible oils (such as cooking and olive oils), and other polymers.

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